

It has become one of the most popular tools in DevOps practices due to its features with respect to deployment, management and scaling applications in Kubernetes clusters.

A few important Kubernetes commands

Description	Command
Get Kubectl version	<code>kubectl version</code>
Get details on Kubernetes cluster	<code>kubectl cluster-info</code>
Get details about a node	<code>kubectl describe nodeName</code>
Get details about a pod	<code>kubectl describe podName</code>
List of all pods	<code>kubectl get pods</code>
List of all services	<code>kubectl get services</code>
List of all deployments	<code>kubectl get deployments</code>

We can create Kubernetes clusters in an on-premise environment and in different cloud deployment models such as private and public cloud. Public cloud provides managed services for Kubernetes. AWS has Amazon Elastic Kubernetes Service (EKS) while Azure has Azure Kubernetes Service (AKS) for managing Kubernetes clusters in a cloud environment.

Terraform

Terraform is an open source 'Infrastructure as Code' tool, created by HashiCorp. It helps us to create immutable infrastructure to avoid the problem of configuration drift.

Terraform characteristics

Open source	Yes
Licence	Mozilla Public License v2.0
Type	Infrastructure as code
Written in language(s)	Go
Operating system supported	Linux, FreeBSD, macOS, OpenBSD, Solaris, and Microsoft Windows
Stable release	0.14.2
Website	https://www.terraform.io/
Companies and projects that use Terraform	Slack, Uber, Twitch, Intuit, CircleCI, Lime

A few important Terraform commands

Description	Command
Verify Terraform version	<code>terraform --version</code>
Install Terraform modules	<code>terraform init</code> <code>terraform init -input=false</code>
Check configuration and create plan	<code>terraform plan</code>
Execute the plan	<code>terraform apply</code> <code>terraform apply -auto-approve</code>
Delete all resources	<code>terraform destroy</code> <code>terraform plan -destroy</code>
Validate/check the syntax of the Terraform files	<code>terraform validate</code>

Terraform is a platform-agnostic declarative coding tool, and helps developers to use a high level configuration language to manage infrastructure. It supports different cloud providers and their services such as AWS, Azure, and so on.

Ansible

Ansible is an open source IT automation/configuration management tool that provides cross-platform support. It is easy to use and offers security and reliability when adopted in an organisation. Ansible is also popular for rolling updates in release management activities.

Ansible characteristics

Open source	Yes
Licence	Proprietary / GNU General Public License
Type	Configuration management
Written in language(s)	Python, PowerShell, Shell, Ruby
Operating system supported	Linux, UNIX-like, MacOS, Windows
Stable release	2.10.2
Website	https://www.ansible.com/
Companies and projects that use Ansible	Atlassian, AWS, Azure, VMware, Google Cloud Platform, Cisco

Ansible is different from other popular configuration management tools because it is agentless. It can be integrated with Kerberos, LDAP, and other authentication management systems.